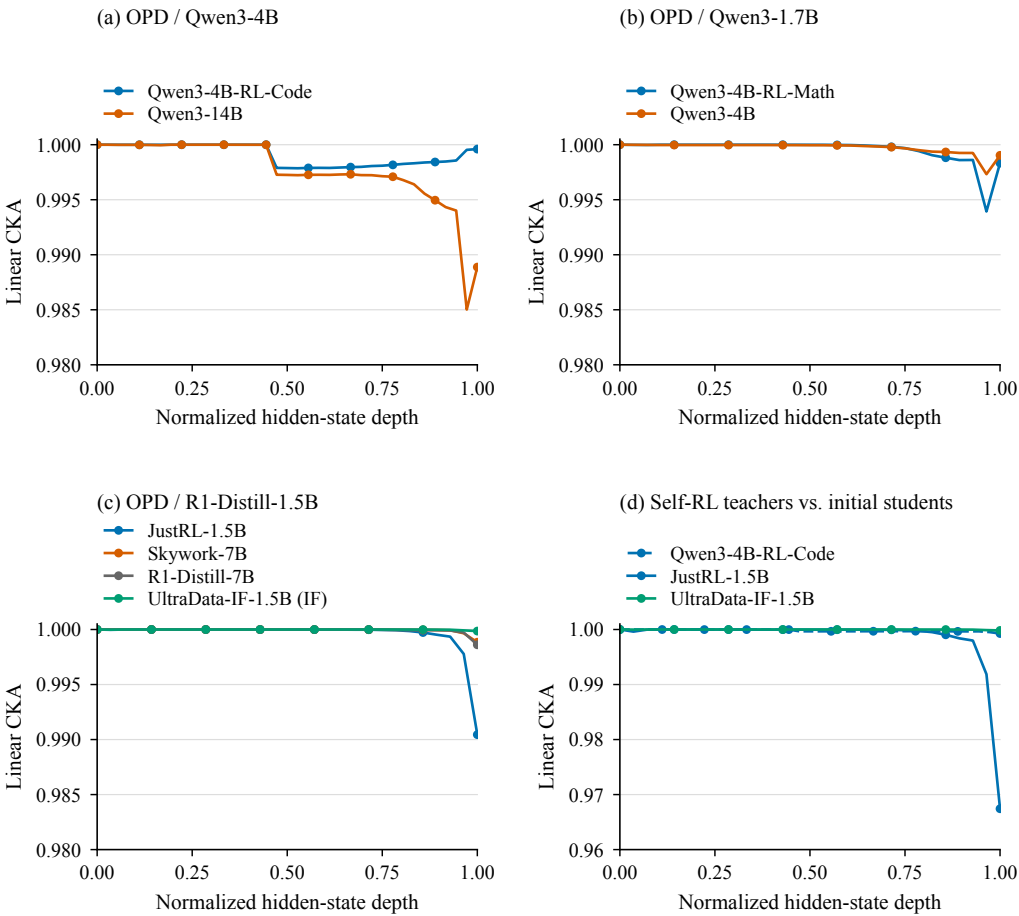
(a) OPD / Qwen3-4B
Qwen3-4B-RL-Code
Qwen3-14B
Linear CKA
1.000
0.995
0.990
0.985
0.980
0.00
0.25
0.50
0.75
1.00
Normalized hidden-state depth
(b) OPD / Qwen3-1.7B
Qwen3-4B-RL-Math
Qwen3-4B
Linear CKA
1.000
0.995
0.990
0.985
0.980
0.00
0.25
0.50
0.75
1.00
Normalized hidden-state depth
(c) OPD / R1-Distill-1.5B
JustRL-1.5B
Skywork-7B
R1-Distill-7B
UltraData-IF-1.5B (IF)
Linear CKA
1.000
0.995
0.990
0.985
0.980
0.00
0.25
0.50
0.75
1.00
Normalized hidden-state depth
(d) Self-RL teachers vs. initial students
Qwen3-4B-RL-Code
JustRL-1.5B
UltraData-IF-1.5B
Linear CKA
1.00
0.99
0.98
0.97
0.96
0.00
0.25
0.50
0.75
1.00
Normalized hidden-state depth